INFORMATION TO KEEP WITH YOU

My complete treatment handbook

CMF · oral cyclophosphamide

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
28 days
See the active treatment phase

Treatment stages
1
Explained in the treatment reference

Regimen source
NCCP 00377
Version 6

> Cyclophosphamide tablets on Days 1–14, with intravenous (IV) methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated.

Treatment stages

01 **CMF · oral cyclophosphamide**
28-day reference

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Contact oncology promptly for fever, shivering, unusual bleeding, severe mouth or bowel symptoms, inability to drink, reduced urine, new breathlessness or a painful/leaking line. Severe chest pain, collapse or severe breathing difficulty needs emergency help.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

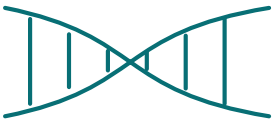
How the medicines work

After surgery: reduce recurrence risk. In metastatic disease: control cancer and symptoms. Your team confirms which setting applies.

ALKYLATING CHEMOTHERAPY

Cyclophosphamide

Tablets on Days 1–14 only



Damages deoxyribonucleic acid (the cell's genetic material) (DNA) needed for growth

Cyclophosphamide is activated in the body and damages DNA. This helps stop dividing cancer cells but can also affect blood-cell production and reproductive tissues. In this regimen it is the only take-home chemotherapy tablet; methotrexate and fluorouracil are given in the unit.

Drug activated → DNA damage → Cell division affected

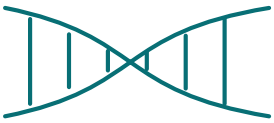
What to watch for

Use the prescribed anti-sickness and hydration plan. Report fever, painful urination or blood in the urine.

ANTIFOLATE CHEMOTHERAPY

Methotrexate

intravenous (IV) injection in CMF



Interferes with folate-dependent DNA production

Methotrexate reduces the supply of building blocks needed to make DNA. Kidney function affects its clearance, and fluid collections can prolong exposure. In CMF it is given intravenously by the unit; it is not a take-home weekly tablet.

Folate pathway blocked → DNA building blocks reduced → Cell growth disrupted

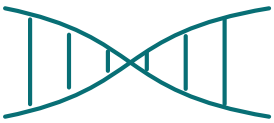
What to watch for

Report mouth ulcers, diarrhoea, poor intake, reduced urine, new cough or breathlessness. Ask before starting medicines or supplements.

ANTIMETABOLITE CHEMOTHERAPY

Fluorouracil (5-FU)

5-FU by IV injection on the specified CMF days



Disrupts DNA and RNA production

Fluorouracil interferes with DNA and RNA production. In CMF it is given by injection in the unit. dihydropyrimidine dehydrogenase (a medicine-processing enzyme) (DPD) testing helps identify increased toxicity risk; it does not exclude all serious reactions.

Cell building blocks disrupted → DNA / RNA affected → Growth reduced

What to watch for

Report severe diarrhoea, mouth ulcers or inability to drink promptly. New chest pain needs urgent assessment.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

CMF is an anthracycline-free chemotherapy option. After surgery it may be considered when an anthracycline is unsuitable; it also has a metastatic indication. The team considers cancer biology, prior treatment, fitness, kidney function and your ability to follow the tablet schedule.

Treatment setting

Adjuvant when anthracyclines are unsuitable, or metastatic breast cancer

Your exact schedule

Cyclophosphamide tablets on Days 1–14, with IV methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated.

Alternatives

Discuss suitable endocrine, targeted or chemotherapy options, expected benefit, practical demands and the option of supportive care according to your setting.

Why this treatment may be chosen

Show trial results and what the numbers mean

Pembrolizumab + chemotherapy

Placebo + chemotherapy

Your team can discuss expected benefit, alternatives and uncertainty for your own cancer.

CHAPTER 02

Your schedule, tests and appointments

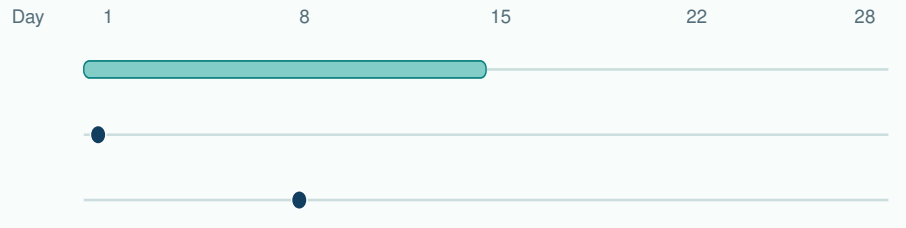
IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

The phases and how they connect

CMF · oral cyclophosphamide · 28-day reference

28-day view



This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

28-DAY REFERENCE

CMF · oral cyclophosphamide

28-day interval

Cyclophosphamide tablets on Days 1–14, with IV methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated.

These are the medicines in this exact schedule. Dose holds, missed tablets and omitted visits need instructions from your treating team.

BASELINE → TREATMENT → FOLLOW-UP

When checks happen

Relative milestones, not equal time intervals. Your team books tests and reviews results; this map does not confirm that either has happened.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00377 v6 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · NICE NG101 · bone health and follow-up · ASCO · biomarkers in metastatic breast cancer · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

CMF · oral cyclophosphamide: treatment and checks

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	First treatment block	During treatment	Change / completion	Follow-up
Cancer medicines NCCP 00377	Confirm exact regimen	CMF · oral cyclophosphamide	Cyclophosphamide tablets on Days 1–14, with IV methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated.	Review active medicines and purpose	Next treatment / follow-up plan
Surgery / radiotherapy NCCP 00377	Confirm adjuvant or metastatic setting	Follow the multidisciplinary team (MDT) sequence	Radiotherapy timing needs agreement	Selected local treatment as indicated	Review recovery and ongoing effects
Blood / kidney / liver NCCP 00377	full blood count (FBC), renal and liver profile	Confirm treatment-day results	FBC, renal and liver tests before each cycle; confirm Day 8 review	Review unresolved abnormalities	Individual clinical plan
Heart / breathing NCCP 00377	Review symptoms and prior treatment	Investigate clinical concerns	No routine HER2-style echo schedule	Review new chest/ breathing symptoms	Clinical indication
Cancer review NCCP 00377 · EARLY · METASTATIC · MARKERS	Confirm stage and setting	Agree review checkpoints	Review symptoms and response	Discuss continuation / next treatment	Setting-specific review
Other checks NCCP 00377 · BONE	DPD before 5-FU	Review oral-dose instructions	Review toxicity before each dose	Endocrine / bone plan where relevant	Follow unresolved symptoms

What each check is for

1



Baseline review

Before treatment

Adjuvant when anthracyclines are unsuitable, or metastatic breast cancer. Confirm DPD testing before the first fluorouracil dose. Kidney impairment can affect interpretation of uracil-based testing; ask the oncology team to review the result and testing method.

Ask / **act:** Confirm source eligibility and results.

2

Treatment-day check

Before each planned dose

NCCP specifies FBC, kidney and liver tests before each cycle. The unit must also confirm the Day 8 assessment and oral-dose instructions.

Ask / **act:** Confirm active medicines and whether a dose is held.

3

Support plan

At every phase or day change

Cyclophosphamide tablets on Days 1–14, with IV methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated.

Ask / **act:** Choose the matching support script; do not merge phases.

4

Response and follow-up

Agreed review points

The treatment setting changes the scan plan. After early-breast treatment, breast imaging and clinical review follow the agreed plan. Routine computed tomography (CT)/positron emission tomography (PET) scans or blood tumour markers are not used to look for distant recurrence in a person without symptoms or clinical concerns. For metastatic disease, record the disease sites and agree the scan type and review interval before treatment. Review response, symptoms and toxicity together; worsening symptoms may need earlier assessment.

Ask / **act:** Review symptoms, results and patient priorities together.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan

STEP 1

Before the first visit

Blood tests, medicines review and the consent discussion come first. Confirm DPD testing before fluorouracil and review methotrexate kidney clearance.

STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

Take the prescribed cyclophosphamide tablets only on Days 1–14. Keep them safely and follow the pharmacy handling advice. Keep the Day 8 IV appointment and confirm the next cycle before restarting tablets.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

Allow time for blood review, preparation, treatment and observation. Your unit confirms the full visit duration. Arrange transport or an escort where advised.

Phase-specific visit pattern

Cyclophosphamide tablets on Days 1–14, with IV methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated. No take-home infusion pump is part of this schedule. Port de-accessing after a visit is different from removal of the implanted port.

Checks across treatment

Your clinical team books tests, reviews results and explains any change. Contact them if an expected test or instruction is missing.

Blood and kidney tests

NCCP specifies FBC, kidney and liver tests before each cycle. The unit must also confirm the Day 8 assessment and oral-dose instructions.

DPD before fluorouracil

Confirm DPD testing before the first fluorouracil dose. Kidney impairment can affect interpretation of uracil-based testing; ask the oncology team to review the result and testing method.

Heart and breathing

There is no routine three-month echo schedule for CMF or gemcitabine alone. Review prior treatment and cardiac risk; new chest pain or breathing symptoms need urgent assessment.

Cancer imaging

The treatment setting changes the scan plan. After early-breast treatment, breast imaging and clinical review follow the agreed plan. Routine CT/PET scans or blood tumour markers are not used to look for distant recurrence in a person without symptoms or clinical concerns. For metastatic disease, record the disease sites and agree the scan type and review interval before treatment. Review response, symptoms and toxicity together; worsening symptoms may need earlier assessment.

Bone health

Bone-density testing follows endocrine treatment, ovarian suppression, menopause and other bone-risk factors. dual-energy X-ray absorptiometry (bone density) (DEXA) is not automatically required before every chemotherapy course.

The wider treatment plan

For adjuvant CMF, surgery has usually already happened; agree any later radiotherapy and endocrine treatment separately. In metastatic disease, surgery or radiotherapy is considered for selected clinical needs, not automatically after a fixed number of cycles.

When treatment changes

Reconcile active cancer medicines, supportive medicines and remaining symptoms before every transition.

Follow-up

Keep the agreed review appointments and report new symptoms between visits; this map does not detect missed tests.

Hormone treatment and bone care

Hormone-receptor results and your treatment setting determine whether endocrine therapy is appropriate. It has its own monitoring plan.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

CMF · oral cyclophosphamide · 28-day reference

Day 1 → Day 28: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Cyclophosphamide tablets	 **Days 1–14** Days 1–14 only
IV treatment · Day 1	 **Day 1** Treatment in the unit
IV treatment · Day 8	 **Day 8** Treatment in the unit
Blood counts / infection	 **Days 1–28** Fever needs urgent advice on any day
Fatigue	 **Days 1–28** May build across cycles
Nausea / appetite	 **Days 1–28** Follow the written prevention plan
Mouth / bowel	 **Days 1–28** Report ulcers, diarrhoea or poor intake

General orientation: Symptoms can occur at any time, including days without a treatment visit. The graph does not predict symptom-free days.

Treatment days: A delayed or omitted dose needs an updated plan from the unit. Do not move doses yourself.

Why these bands stay broad: These broad symptom bands are educational possibilities, not measured risk curves. Infusion days are the planned schedule; delays require a new clinical plan. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00377 v6 · official regimen ↗

There are no guaranteed "safe days". Fever, shivers, sudden illness, severe breathing changes or chest pain need prompt assessment on any day.

Between-treatment notes

Day 1–28: One reference cycle

Cyclophosphamide tablets on Days 1–14, with IV methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated. Use your written appointment and medicine plan; no dates are entered here.

Support around the medicines

1 **Tablet days and safe handling**

Cyclophosphamide is taken on Days 1–14 only. Swallow tablets whole, one hour before food or on an empty stomach, as prescribed. Ask the unit what to do after a missed or vomited dose; do not double a dose. Methotrexate is given by the unit, not taken at home.

2 **Anti-sickness plan**

Follow the prevention and rescue instructions prescribed for this schedule. Call if vomiting prevents drinking.

3 **Infection and bleeding**

Know the local urgent-contact number. Fever, shivering, sudden illness or unusual bleeding needs prompt advice even between visits.

4 **Medicine review**

Show the oncology pharmacist all prescribed, over-the-counter and herbal medicines. Ask before starting anything new.

5 **Hydration and symptoms**

Agree fluid intake with the team, especially with heart or kidney fluid restrictions. Report reduced urine, swelling, mouth ulcers or diarrhoea.

6 DPD result and methotrexate clearance

Confirm DPD testing before the first fluorouracil dose. Kidney impairment can affect interpretation of uracil-based testing; ask the oncology team to review the result and testing method. Tell the team about fluid around the lungs or in the abdomen, which can prolong methotrexate exposure.

Questions worth asking

Who arranges the next tests?

Your treating service arranges and reviews tests. If instructions or appointments are missing, contact the team; the map does not book tests.

What if a dose is missed or I feel unwell?

Contact the unit for a revised plan. Do not double tablets, shift treatment dates or assume that an omitted dose should be made up.

Try explaining the plan back: Which medicines are due on which days? What symptoms mean you should call today? Which tests come next and who arranges them?

Cyclophosphamide

- Low counts and infection
- Nausea, vomiting and hair loss
- Bladder irritation / bleeding
- Fertility effects and menopause
- Rare later blood cancers

Methotrexate

- Low blood counts and serious infection
- Mouth ulcers and diarrhoea
- Liver injury and prolonged exposure with kidney impairment
- Uncommon serious lung inflammation

Fluorouracil (5-FU)

- Low blood counts and infection
- Diarrhoea and mouth ulcers
- Rare serious heart symptoms
- Severe toxicity with dihydropyrimidine dehydrogenase (a medicine-processing enzyme) (DPD) deficiency

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Days 1–14 · oral cyclophosphamide plus intravenous (IV) visits

Active medicines: Cyclophosphamide tablets Days 1–14 + Methotrexate and fluorouracil IV Days 1 and 8

Emetogenic risk: Moderate–high oral cyclophosphamide; IV agents low

Medicine	Dose / route	When
Ondansetron	8–16 mg total orally per day; prescriber selects exact dose	Start before systemic anticancer therapy (SACT) and continue daily through the oral-treatment course; confirm stop day NCCP oral Table 5. Review QT, constipation and hepatic dosing. Reconcile the same daily total on IV Days 1 and 8; do not add a duplicate IV-day order.

Additional phase-specific support

- Confirm allergy/interaction review, blood tests and the source-specific renal/hepatic assessment before prescribing.
- Confirm dihydropyrimidine dehydrogenase (a medicine-processing enzyme) (DPD) status before first fluorouracil and review partial deficiency with the consultant.
- Confirm an individual hydration and bladder-symptom plan; adapt fluid advice to cardiac/renal restrictions.
- Review pleural effusion or ascites because methotrexate can accumulate in third-space fluid.
- NCCP oral guidance uses a daily 5-HT3 antagonist. Do not automatically apply the single-day parenteral high-risk/AC steroid schedule to this 14-day oral course.
- Specify rescue medicine and duration, and reconcile prophylaxis for two days after multiday SACT under the local plan.
- Cyclophosphamide tablets are Days 1–14 only; give clear missed/vomited-dose instructions. Methotrexate is IV in this regimen.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

CMF · oral cyclophosphamide

Cyclophosphamide tablets on Days 1–14, with intravenous (IV) methotrexate and fluorouracil on Days 1 and 8 of each 28-day cycle. Six cycles for adjuvant treatment; metastatic treatment continues while beneficial and tolerated.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.

1 Mouth

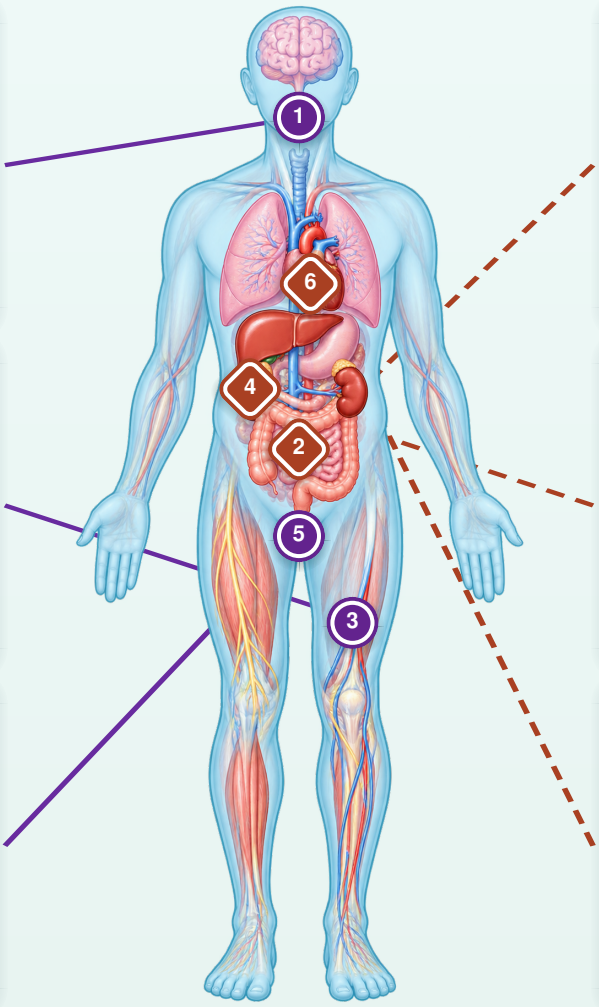
Methotrexate / fluorouracil

Ulcers or difficulty drinking need prompt advice.

2 Bowel

Methotrexate / fluorouracil

Report diarrhoea, vomiting or poor intake.



3 Blood / infection

All three medicines

Fever, shivering or sudden illness: call immediately.

4 Kidneys / fluid

Methotrexate

Kidney function affects clearance; report reduced urine.

5 Bladder

Cyclophosphamide

Report painful urination or blood in urine.

6 Chest / breathing

Fluorouracil / methotrexate

New chest pain or breathlessness needs urgent assessment.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Medicine-specific effects are highlighted; this is not a complete list. New or persistent symptoms still need review after treatment stops.

Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00377 v6 · official regimen**
   https://healthservice.hse.ie/documents/6337/377_v6_CMF_oral.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
4. **ASCO · biomarkers in metastatic breast cancer**
   https://ascopubs.org/doi/10.1200/JCO.2015.61.1459
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
7. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
8. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
9. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
10. **NOGG · section 7, aromatase inhibitors and fracture risk**
    https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
11. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
12. **NCCP antiemetic classification v6 · oral therapy Table 5, p8**
    https://assets.hse.ie/media/documents/Classification_Document_for_Systemic_Anti-_Cancer_Therapy_systemic anticancer therapy (SACT)_Induced_Nausea__pWgDCrm.pdf